Title: Global Needs

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Introduction about Global Needs



AI Water Energy

- The growing demand for AI and cloud computing increases electricity consumption and water usage worldwide.
- Efficient resource management is essential to reduce wastage and promote sustainable computing.
- Global need: Predict future electricity and water needs for data centers to enable green, sustainable growth.

PROBLEM STATEMENT

Digital Resource Management: The Energy–Compute Balance





Brief Overview of the Project: Digital Resource Management - The Energy-Compute Balance

- This project focuses on improving energy and water efficiency in AI data centers using Machine Learning (ML) techniques.
- As AI workloads increase, data centers require more electricity for computation and water for cooling.
- The proposed system uses historical resource consumption data and the Random Forest algorithm to predict future resource requirements and identify opportunities to reduce resource wastage.
- The main objective is to optimize energy and water usage while maintaining computing performance and supporting sustainable digital resource management.


Server


Water


Energy


ML Brain


Tree

PROBLEM ADDRESSED - Digital Resource Management: The Energy-Compute Balance

1. AI data centers consume excessive electricity for computing tasks.
2. Cooling systems increase energy consumption and water usage.
3. Increasing workloads make resource management more difficult.
4. Traditional methods may not accurately predict future resource requirements.
5. Machine Learning helps predict consumption and reduce resource wastage.



Proposed solution:Uses historical data and the Random Forest algorithm to predict electricity and water consumption in data centers.

Recommends efficient workload scheduling and cooling strategies to reduce resource wastage while maintaining performance.



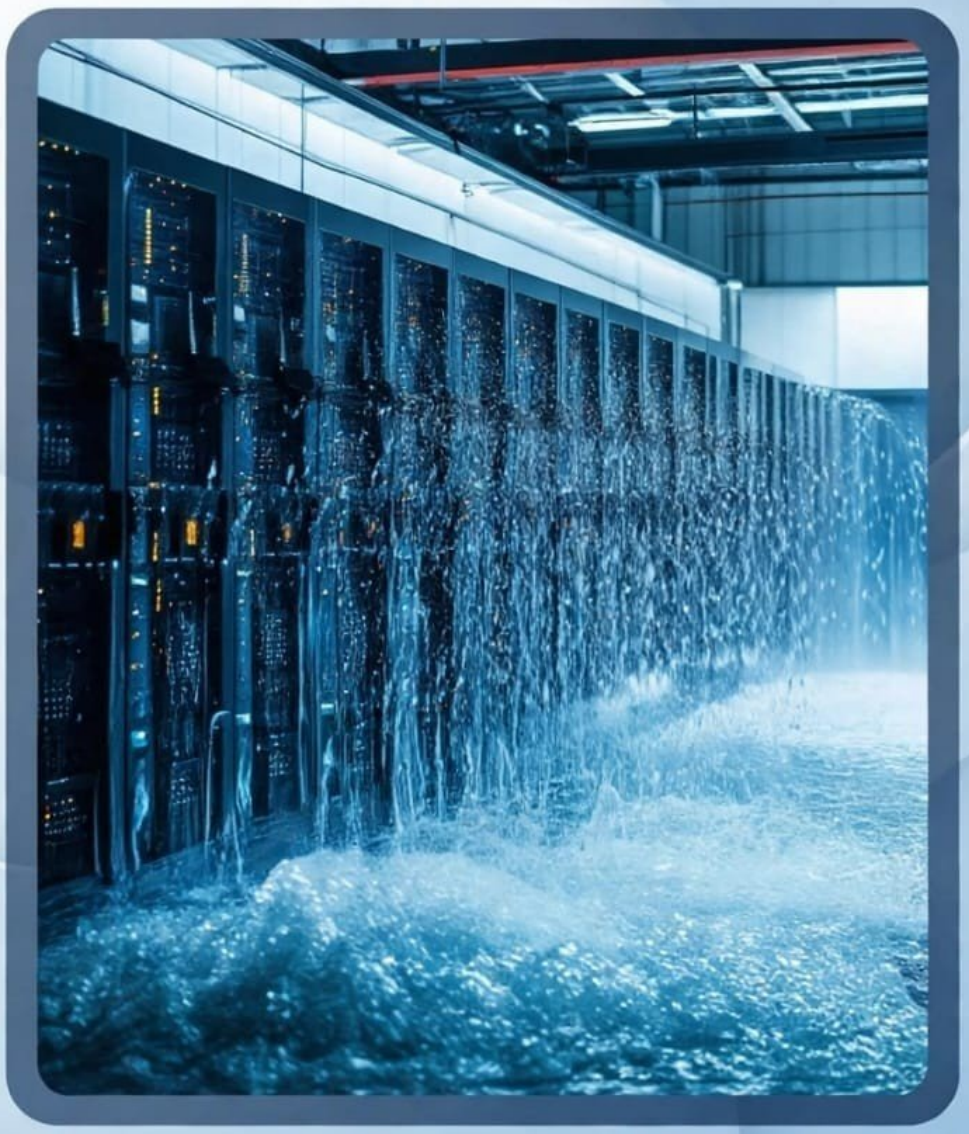
Unique feature:Uses the Random Forest algorithm to predict electricity and water consumption in AI data centers.

Identifies resource wastage and recommends efficient resource management for sustainable computing.



Introduction

- **AI & Cloud growth** → high demand for data center resources.
- Data centers consume high electricity & water for cooling.
- Leads to high costs & environmental impact for operators & businesses.
- Solution: ML with Random Forest Regression to predict electricity & water usage for sustainable management.



⚠ Critical Issue: Water Consumption

- 💧 AI data centres could consume water equivalent to 1.3 billion people
- 💧 Huge cooling requirements lead to massive water wastage
- 💧 Increases operational cost and environmental impact
- 💧 Need for ML-based prediction for sustainable management

Literature Survey – Introduction

Title: Digital Resource Management: The Energy-Compute Balance

AI data centers require electricity for computing and water for cooling.

Increasing workloads increase resource consumption.

Machine Learning can help predict future resource requirements.

Random Forest Regression is used to predict continuous values.

This project explores energy and water consumption prediction.



How the Existing System Works

CPU Room
Temperature & Humidity

Server Rack
Temperature & Door Status

Power Facility
Operational Status

Cooling Facility
Operational Status

Additional Measurement
Monitoring

Step-by-Step Process

1 Step 1: Data Collection

- The system collects information about electricity consumption, water usage, temperature, and server workload.

2 Step 2: Resource Monitoring

- Monitoring tools track the energy consumed by servers and cooling systems.

How the Existing System Works - Continued

Step 3: Fixed Rules

The system uses predefined rules and thresholds to determine when cooling or other resources need adjustment.

Step 4: Cooling Management

Cooling systems are adjusted to maintain the required temperature and prevent servers from overheating.

Step 5: Resource Usage Analysis

Historical reports are reviewed to identify high consumption and possible wastage.

Step 6: Manual Adjustment

Staff may adjust server workloads and cooling settings based on monitoring results.

Methodology: Historical Data Collection

Style B
Data Center Tech

Year	Electricity (MWh)	Water (Million Gallons)
2020	10,000	3,749
2021	11,000	4,562
2022	12,500	5,000
2023	14,000	5,500
2024	15,500	6,000
2025	17,000	6,500

Trend: +70% Electricity Growth • +73.5% Water Use Growth (2020–2025)

Bootstrap Sampling (Bagging)

Random Forest creates multiple training datasets by randomly selecting rows from the original dataset, with replacement.

This process is called bootstrap sampling.

For example:

Sample A

Electricity (MWh)	Water (Million Gallons)
10,000	3,749
11,000	4,562
12,500	5,000
14,000	5,500

Sample B

Electricity (MWh)	Water (Million Gallons)
11,000	4,562
12,500	5,000
14,000	5,500
17,000	6,500

Electricity (MWh)	Water (Million Gallons)
10,000	3,749
12,500	5,000
15,500	6,000
17,000	6,500

Explanation:

- Each sample is used to train a different decision tree.
- A row may appear more than once in a bootstrap sample.
- Different samples help create diversity among the trees.

STEP 3: Train Individual Decision Trees

Each decision tree learns rules from its training sample. For regression, a decision tree divides the data into groups and predicts the average water consumption of the training observations in the final leaf.

Suppose the trained trees produce these example predictions:

Decision Tree	Example Water Prediction
Tree A	4,562 million gallons
Tree B	5,000 million gallons
Tree C	5,500 million gallons

STEP 4: Ensemble Aggregation (Averaging)

Random Forest combines the predictions from all the decision trees.
Your example uses:

Tree	Water Prediction
Tree A	4,562
Tree B	5,000
Tree C	5,500

Step 1: Add the predictions.

4562 + 5000 + 5500 = 15062

Step 2: Divide by the number of trees. There are three trees.

Final Prediction = 15062 / 3 = 5020.67 million gallons

Step 2: Divide by the number of trees. There are three trees.

Final Prediction = 15062 / 3 = 5020.6667

Rounded to the nearest whole number: 5,021 million gallons

This is how the example arrives at 5,021 million gallons.

Strictly speaking, the unrounded average is 5,020.67, which rounds to 5,021.

STEP 5: Final Prediction - What Does It Mean?

- The model's example output is approximately 5,021 million gallons of water. However, there is an important distinction: the arithmetic average is calculated correctly.
- The three tree predictions are illustrative assumptions, not outputs from a trained Random Forest model. Therefore, 5,021 million gallons is an example calculation, not a verified prediction of actual water consumption.

- To make a genuine prediction, we must train the Random Forest model on the historical data and supply a new electricity consumption value. Also, the dataset contains only six observations, so it is too small to establish reliable real-world forecasting accuracy.

Proposed Solution —
How the Proposed System Works (1/3)



Step 1: Data Collection

- Collect historical data on electricity consumption
- Collect historical data on water usage
- Collect data on server workload, CPU utilization, and temperature

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Step 2: Data Preprocessing

- Clean the dataset to remove inconsistencies and duplicates
- Handle missing values via imputation/interpolation
- Prepare the data for machine learning analysis

Proposed Solution - How the Proposed System Works (2/3)



Step 3: Apply Random Forest Regression

- Train a Random Forest model using historical data
- Predict electricity consumption and water requirements



Step 4: Predict Resource Consumption

- Estimate future electricity and water usage
- Based on workload and operating conditions

Model trained on historical operational data • Outputs validated for infrastructure planning

Proposed Solution — How the Proposed System Works (3/3)



Step 5: Resource Optimization

- Use the predictions to recommend efficient workload scheduling and cooling adjustments
  - Optimize server loads to balance performance & power usage
  - Adjust cooling systems dynamically to reduce energy waste



Step 6: Evaluate Results

- Compare predicted consumption with actual consumption using metrics such as

MAE	RMSE	R^2
Mean Absolute Error	Root Mean Square Error	Coefficient of Determination

- Assess model accuracy to ensure reliable energy forecasts for operations

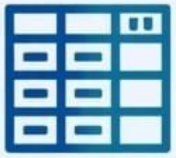
Continuous feedback loop → improves predictions over time

3. Technologies and Algorithms Used



Python

Programming language



Pandas and NumPy

Data processing and calculations



Scikit-learn

Machine Learning implementation



Random Forest Regression

Resource consumption prediction



Matplotlib

Graphs and visualization

5. Advantages & Expected Outcome

Advantages:

- Improved resource-consumption prediction
- Better energy and water management
- Data-driven decision-making
- Potential reduction in resource wastage
- Supports sustainable and efficient data center operations

Expected Outcome:

Predict resource requirements and support efficient energy and water management while maintaining computing performance.


Thank you Rajalakshmi Madam